Wavelet Decomposition-Based Spectral–Spatial Mamba Network for Hyperspectral Image Classification

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***Abstract*— Existing hyperspectral image classification (HSIC) models based on the Mamba architecture predominantly center on characterizing the original spectral and spatial domains, with limited exploration of time–frequency analysis. In this study, we propose a novel wavelet decomposition-based spectral–spatial Mamba network for HSIC, dubbed "WD-SSMamba." This model incorporates both 1-D and 2-D wavelet decompositions to extract spectral and spatial features in the frequency domain, respectively. Specifically, we design an innovative frequency feature extraction (FE) block, which comprises a spectral wavelet convolution (SWC) module for spectral FE and a wavelet separable convolution (WSC) module for spatial FE. Furthermore, to address the challenges of integrating spectral–spatial features in traditional Mamba models for HSIC, we devise a dual-branch Mamba block featuring a cross-fusion structure, termed the "HyperMamba" block, to efficiently extract and fuse spectral and spatial features. Comprehensive experiments were conducted on four publicly available hyperspectral image (HSI) datasets, namely, Pavia University, WHU-Hi-LongKou, WHU-Hi-HongHu, and Houston 2013. The results show that the WD-SSMamba model achieves overall accuracies (OAs) of 92.48%, 97.91%, 89.98%, and 87.00% on these datasets, respectively, with fewer than 20 training samples per class, surpassing those of other competing models across all tested datasets. Moreover, it significantly reduces the number of parameters to less than 50k and floating-point operations (FLOPs) to less than 3.02M, thereby fully showcasing the immense potential of frequency analysis and the Mamba structure in HSIC.**

***Index Terms*— Hyperspectral image classification (HSIC), Mamba, spectral–spatial feature, wavelet decomposition.**

I. INTRODUCTION

NOWADAYS, hyperspectral imaging technology is becoming a significant research topic in the field of remote sensing, due to its ability to capture fine spectral information of ground objects. Many applications, such as environmental monitoring [1], precision agriculture [2], urban planning [3], resource exploration [4], and military security [5], have demonstrated great potential of hyperspectral imaging technology. The hyperspectral image classification (HSIC), which aims to assign a unique land-cover category to each pixel, is the most crucial step in many of the applications mentioned above.

With the rapid development of deep learning, convolutional neural networks (CNNs) and Transformers have shown growing use in HSIC [6], [7], [8], [9], [10], [11], [12], [13], [14], [15]. However, CNNs are limited by their local receptive field, making it difficult to effectively capture global information [16]. Relatively speaking, Transformers excel at modeling long-range dependencies to capture global information through a self-attention mechanism; their computational complexity is $O(n^2 \cdot d)$, where n represents the sequence length and d is the dimension of each token [17]. When dealing with hyperspectral image (HSI) data [18], the input image should be divided into local patches (e.g., 16×16 patches), and n will still reach 256. In this case, although the dimension can be reduced via embedding techniques, the computational cost still scales quadratically with n^2.

To address these challenges, the Mamba model, built upon the state space model (SSM) [19], has emerged as an innovative sequence modeling approach [20]. Mamba can dynamically capture contextual dependencies within sequential data, enabling efficient global feature modeling. Moreover, Mamba can achieve linear or near-linear scaling with sequence length by using a convolutional-style training operation and recurrent-style inference operation, thereby significantly reducing computational costs while maintaining model efficacy [21]. In this regard, Mamba structures have been introduced into the field of vision as a foundational model [22], [23], and increasingly demonstrated their advantages in dealing with complex visual tasks [24] also covering HSIC. MambaHSI [25] and SS-Mamba [26] both adopted multiple cascaded Mamba blocks, and in each Mamba block, a parallel dual-branch architecture for spectral and spatial feature extraction (FE) was constructed. The difference between them is that MambaHSI used independent pixel information to generate tokens, while SS-Mamba used spatial patches to generate tokens. This dual-branch architecture lays the

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foundation for the preferred design of Mamba blocks suitable for HSIs.

Considering that the selective scan mechanism in traditional Mamba is confined to modeling causal sequences, with the limitations of 1-D and a single scanning direction. When it is used for HSIC, one research idea is to perfect the scanning mechanism to fit for hyperspectral high-dimensional data. Yao et al. [27] constructed a piecewise sequential scanning mechanism to model the correlations between spectral bands. Wang et al. [28] introduced the S^2Mamba model, in which a patch-cross scanning mechanism is employed to capture spatial correlation between pixels, and a bidirectional spectral scanning mechanism to capture correlations between spectral bands. Furthermore, in 3DSS-Mamba [29], a 3-D spectral–spatial selective scanning (3DSS) mechanism was customized to perform pixelwise selective scanning on 3-D spectral–spatial tokens.

On the other hand, only relying on the global features provided by Mamba, while ignoring the local homogeneity of ground objects, is easy to cause fragmentation of classification maps, which leads to another research line of integrating global and local features [30]. In SpectralMamba [27], the local spectral–spatial features are extracted by combining depthwise convolution and pointwise convolution; subsequently, piecewise sequential scanning is performed on these merged spectra. In DualMamba [31], a lightweight spectral–spatial residual convolution module is adopted to extract local spectral–spatial features; then, global Mamba features and local convolution features are fused for classification.

It should be noted, the existing studies on Mamba have not considered the FE in the frequency domain. As declared in [32], the frequency-domain features of spectral signature contain more subtle variations between spectral bands, which will significantly facilitate the distinction of ground objects, especially those with spectral similarities. Also in [33], the spatial-domain high- and low-frequency features show finer and smoother discriminant information, which is crucial for understanding complex scenes. However, the extraction of frequency features mentioned above commonly used the fast Fourier transform (FFT). Considering that the basis function of the Fourier transform is a global basis, lacking the ability of time–frequency localization, the FFT cannot meet the frequency requirements of nonstationary signals such as HSIs. In contrast, wavelet transform uses a wavelet basis with finite length and attenuation, which has better time–frequency localization, and is more suitable for processing nonstationary signals. Through wavelet decomposition of HSI, the underlying features can be transformed into different wavelet coefficients, which is conducive to fetching more abundant features.

To this end, we introduce wavelet decomposition to enhance the extraction of spectral–spatial features in the frequency domain. We use a 1-D wavelet transform to decompose the spectral features, separating the high-frequency and low-frequency components from the spectral signature, thereby effectively capturing more subtle variations. Concurrently, we use a 2-D wavelet transform to employ spatial features, enabling the analysis of the spatial structures at different

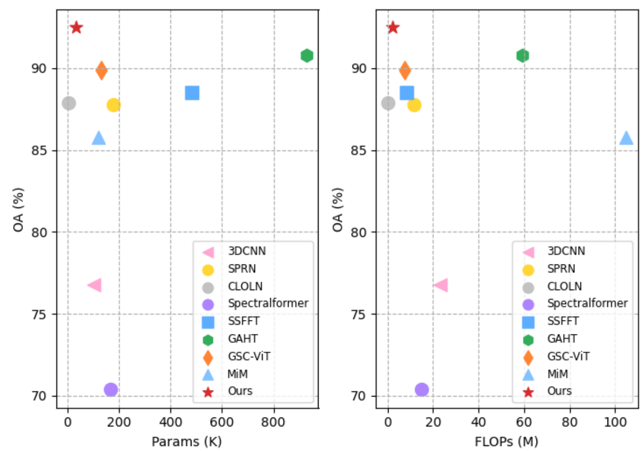


Fig. 1. Performance comparisons with respect to parameters and FLOPs on Pavia University dataset.

frequency levels, thus improving the model's perception of complex scenes. Moreover, in order to enhance the sequence modeling and the fusion abilities of global and local features of Mamba model, as well as optimize the efficient integration of spectral and spatial features, we propose a wavelet decomposition-based spectral–spatial Mamba network for HSIC and use the time–frequency localization characteristics of wavelet transform to complement Mamba's dynamic sequence modeling capabilities, enabling more precise feature representation under limited computational resources. In a nutshell, our work includes three main innovative contributions.

1) We design a spectral wavelet convolution (SWC) module to extract spectral features. SWC module integrates 1-D wavelet transform with convolution operations, decomposing the spectral signature into high- and low-frequency components, thereby delving deeper into the spectrum characteristics from the perspective of the frequency domain.
2) We design a wavelet separable convolution (WSC) module to extract spatial features. WSC module combines 2-D wavelet transform with pointwise convolution, decomposing spatial representation into different frequency components while maintaining high computational efficiency.
3) We construct a "HyperMamba" block, comprising parallel spectral and spatial Mamba branches. Each branch processes features separately in spectral and spatial dimensions, and a cross-fusion framework is built to effectively fuse spectral and spatial features by modeling long-range dependencies through SSM.

In the experiment section, we conducted extensive validation of the proposed network (hereinafter dubbed "WD-SSMamba") on four publicly available HSI datasets. The experimental results demonstrated that our WD-SSMamba model significantly reduces the number of parameters and floating-point operations (FLOPs) while maintaining promising classification accuracies. The potential and applicability of our model for HSIC tasks are validated, as shown in Fig. 1.

The remainder of this article is organized as follows. Section II presents the preliminary work. Section III provides a detailed description of the model architecture and methodology. Section IV presents the experimental design and results. Finally, Section V concludes this article with some remarks and hints at plausible future research lines.

II. Preliminary Work

A. Combination of Wavelet Transform and Deep Learning

Wavelet transform is a versatile tool for analyzing signals in time–frequency domain, adept at extracting frequency-domain features. In recent years, the combination of wavelet and deep learning has shown immense potential across numerous visual tasks [34], [35], [36]. In these endeavors, wavelet is leveraged to complement missing multiresolution analysis and seamlessly incorporated into deep network architectures.

In the realm of HSI, wavelet is extensively utilized for denoising purposes [37], [38], as well as for enhancing spectral and spatial feature representation [39]. Ullah et al. [40] utilized multilayer wavelet transform to extract spectral features, and integrated them into a 2-D CNN. Guo et al. [41] concatenated four-level Haar wavelet decomposition features with four-layer convolution features to obtain sufficient spectral–spatial feature representation. Vatsavayi et al. [42] exploited two different types of discrete wavelet transform (DWT), including Haar and Daubechies (Db4), for spectral FE. As shown in Fig. 2(a), for a 1-D signal $f[n]$, convolving it with low-pass and high-pass filters yields the approximation coefficients (cA) and detail coefficients (cD). cA represents the low-frequency component of the signal, reflecting its overall trend, while cD captures the high-frequency components, highlighting details and variations within the signal. The specific equations are listed in (1). In the context, n denotes the discrete index of the input 1-D signal $f[n]$, corresponding to the spatial or temporal positions of the sampled data points in the original signal, while k represents the index of the wavelet coefficients after decomposition

$$\mathrm{cA}[k] = \sum_n f[n]\cdot\phi[2k-1]$$
$$\mathrm{cD}[k] = \sum_n f[n]\cdot\psi[2k-1]. \tag{1}$$

In (1), $\phi[\cdot]$ is the scaling function used to preserve low-frequency information of the signal, and $\psi[\cdot]$ is the discrete wavelet function employed to capture high-frequency components. By decomposing a spectral signature into high- and low-frequency components, the 1-D wavelet transform effectively captures subtle variations within the spectrum, especially the intricate information across various bands.

In order to fit the HSI data cube, 2-D/3-D DWT is also introduced [43]. Xu et al. [44] adopted 3-D DWT to extract joint spectral–spatial frequency features, and the obtained patch-level wavelet features were fed into a dense CNN. In [45], each image patch was decomposed and reconstructed by wavelet multilevel transform, and the obtained multiscale wavelet spectral–spatial features were input into a Transformer network. Specifically, the 2-D Haar wavelet transform employs

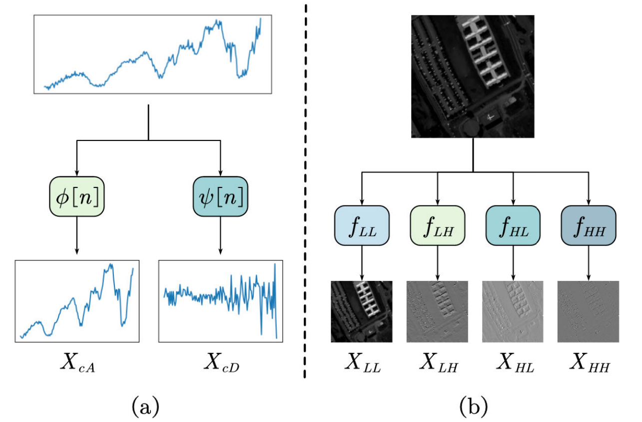


Fig. 2. (a) Schematic of 1-D Haar wavelet decomposition. (b) Schematic of 2-D Haar wavelet decomposition.

low-pass and high-pass filters to decompose an image in horizontal and vertical directions, respectively. As shown in (2) and (3), 2-D Haar wavelet transform can be represented through four convolutional kernels to each channel with a stride of 2, resulting in the division of an image into four subimages: one low-frequency component (LL) and three high-frequency components (LH, HL, and HH), corresponding to approximation information, horizontal details, vertical details, and diagonal details, respectively [as illustrated in Fig. 2(b)]

$$f_{\mathrm{LL}} = \frac{1}{2}\begin{bmatrix}1 & 1\\ 1 & 1\end{bmatrix},\ f_{\mathrm{LH}} = \frac{1}{2}\begin{bmatrix}1 & -1\\ 1 & -1\end{bmatrix}$$
$$f_{\mathrm{HL}} = \frac{1}{2}\begin{bmatrix}1 & 1\\ -1 & -1\end{bmatrix},\ f_{\mathrm{HH}} = \frac{1}{2}\begin{bmatrix}1 & -1\\ -1 & 1\end{bmatrix} \tag{2}$$
$$\begin{bmatrix}X_{\mathrm{LL}} & X_{\mathrm{LH}}\\ X_{\mathrm{HL}} & X_{\mathrm{HH}}\end{bmatrix} = \mathrm{Conv}\left(\begin{bmatrix}f_{\mathrm{LL}} & f_{\mathrm{LH}}\\ f_{\mathrm{HL}} & f_{\mathrm{HH}}\end{bmatrix}, X\right). \tag{3}$$

In order to leverage the advantages of wavelet transform in the frequency-domain FE, we propose a novel spectral–spatial FE block that combines 1-D and 2-D wavelet transforms with convolution.

B. SSM and Mamba

The SSM originates from control theory and has shown great potential in the field of deep learning in recent years, particularly for sequence modeling tasks. SSM describes the relationship between the input sequence $x(t) \in \mathbb{R}$ and the output sequence $y(t) \in \mathbb{R}$ through a hidden state $h(t) \in \mathbb{R}^N$, as shown in the following equation:

$$h'(t) = \mathbf{A}h(t) + \mathbf{B}x(t)$$
$$y(t) = \mathbf{C}h(t) \tag{4}$$

where $\mathbf{A} \in \mathbb{R}^{N\times N}$ is the state transition matrix, describing how the system state evolves over time; and $\mathbf{B} \in \mathbb{R}^{N\times 1}$ and $\mathbf{C} \in \mathbb{R}^{1\times N}$ represent the mappings from input to hidden state and from hidden state to output, respectively.

When dealing with discrete sequence data, the zero-order hold (ZOH) technique is typically employed to discretize the continuous-time model, resulting in the discretized SSM in the

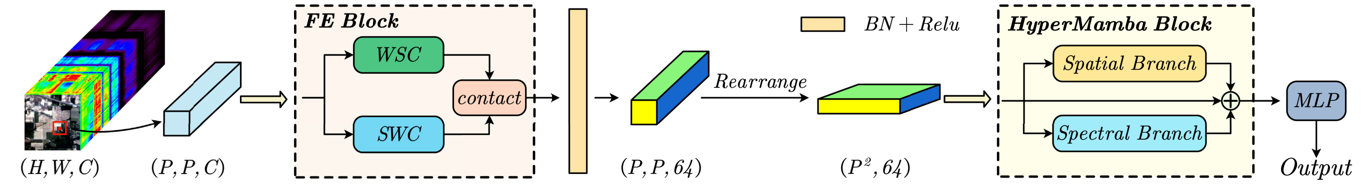


Fig. 3. Schematic of the proposed WD-SSMamba network. "WSC" is the wavelet separable convolution module used for extracting spatial features. "SWC" is the spectral wavelet convolution module used for extracting spectral features. WSC and SWC both have a size of $P \times P \times 32$. "FE block" concatenates the outputs of WSC and SWC, in FE block, "contact" refers to the concatenation operation along the channel dimension, thus generating the final feature map of size $P \times P \times 64$. "HyperMamba block" performs spectral–spatial feature fusion, "MLP" is a multilayer perceptron layer used for final classification.

following equation:

$$\begin{aligned} h_t &= \overline{\mathbf{A}} h_{t-1} + \overline{\mathbf{B}} x_t \\ y_t &= \mathbf{C} h_t. \end{aligned} \tag{5}$$

In (5), the discrete-time parameters $\overline{\mathbf{A}}$ and $\overline{\mathbf{B}}$ and timescale Δ are computed by the following equation:

$$\begin{aligned} \overline{\mathbf{A}} &= e^{\Delta \mathbf{A}} \\ \overline{\mathbf{B}} &= (\Delta \mathbf{A})^{-1} \left(e^{\Delta \mathbf{A}} - I \right) \cdot \Delta \mathbf{B}. \end{aligned} \tag{6}$$

Moreover, the output of SSM can also be represented in the form of a convolution, as shown in the following equation:

$$\begin{aligned} \overline{\mathbf{K}} &= \left(\mathbf{C}\overline{\mathbf{B}}, \mathbf{C}\overline{\mathbf{A}}\overline{\mathbf{B}}, \ldots, \mathbf{C}\overline{\mathbf{A}}^{L-1}\overline{\mathbf{B}} \right) \\ y &= x * \overline{\mathbf{K}} \end{aligned} \tag{7}$$

where L denotes the length of the input sequence x, and $\overline{\mathbf{K}} \in \mathbb{R}^L$ serves as the convolutional kernel. This representation allows us to efficiently compute the sequence output using convolution operations, thereby providing greater flexibility and computational efficiency.

However, the parameters of the SSM are time-invariant, posing certain limitations when dealing with complex dynamic changes. To overcome this issue, the selective SSM (S6) was introduced [20]. S6 enhances the model's adaptability to dynamic changes by dynamically adjusting its key parameters as functions of the input sequence x. This dynamic selection mechanism enables S6 to capture complex temporal dynamics and effectively model global information. Furthermore, the Mamba architecture not only improves the computational efficiency of SSM to near-linear time complexity but also enhances its modeling capability for long sequences [20].

III. Proposed Methodology

This section provides a detailed explanation of our proposed WD-SSMamba model. First, we outline the overall framework and workflow of the model. Next, we delve into the design motivations and implementation of some key components within the model.

A. Overall Framework of WD-SSMamba

Fig. 3 depicts a schematic of the proposed model. At the input stage, the image is split into smaller patches, each with dimensions of $P \times P \times C$, to fully exploit the spatial information surrounding each pixel. These patches are then fed into the FE block, which comprises two parallel submodules: the WSC module and the SWC module. The WSC module integrates a 2-D wavelet transform with convolution to capture spatial structure information in the frequency domain. Concurrently, the SWC module combines a 1-D wavelet transform with convolution to discern subtle spectral variations, thereby enhancing the model's capability to distinguish various land-cover categories. After obtaining the outputs from both WSC and SWC modules, the spectral and spatial features are fused through concatenation, leading to a more refined feature representation. In contrast to the 3-D wavelet transform, the integration of 1-D and 2-D wavelet transforms markedly reduces computational complexity, thereby enabling the model to achieve both high performance and high efficiency. Specifically, the 1-D wavelet transform is designed to extract spectral features, whereas the 2-D wavelet transform is tailored for spatial FE. This targeted approach to FE facilitates subsequent processing within the HyperMamba block. Subsequently, following a rearrangement operation, the fused features are fed into the customized HyperMamba block, where the Mamba architecture is utilized to capture long-range dependencies within the data while preserving the integrity of the features through residual connections. The HyperMamba block is structured with two distinct branches: the spatial Mamba branch and the spectral Mamba branch. Each branch handles the fused features from unique perspectives of spatial and spectral, respectively. At the output stage, the spectral, spatial, and original input features are once again fused together via residual connections, and the multilevel output features are then fed into a multilayer perceptron (MLP) for classification purposes. In addition, to ensure computational efficiency when processing complex hyperspectral data, as well as accommodate the inherent nonlinearity of the data, we also add batch normalization and ReLU activation in several critical layers.

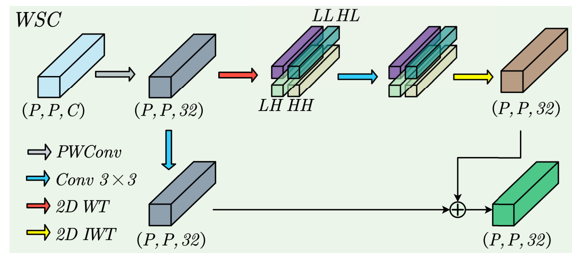


Fig. 4. Spatial FE process of WSC.

B. Wavelet Separable Convolution

As shown in Fig. 4, the WSC module integrates a 2-D wavelet transform with pointwise convolution, enabling the efficient decomposition of spatial frequency-domain information while preserving computational efficiency.

First, the input image patch, denoted as $X_{\text{input}} \in \mathbb{R}^{P\times P\times C}$, undergoes dimensionality reduction through a pointwise convolution layer, which provides optimized feature maps for the next wavelet convolution operations. Subsequently, a 2-D Haar wavelet transform is applied to the dimensionality-reduced feature maps, enabling multiband decomposition across different frequency scales. This results in four subband components: X_{LL}, X_{LH}, X_{HL}, X_{HH}. Each of the decomposed frequency subbands is then processed by a convolution layer, utilizing a convolutional kernel W_1 with a size of 3×3 to effectively extract local refined features of the image while preserving spatial resolution. By following, the feature maps are restored to their initial spatial dimensions through a 2-D inverse wavelet transform, with the corresponding calculation formula shown in (8). In addition, the output of the WSC module employs a customized residual connection, in which the dimensionality-reduced feature maps are convolved with another set of convolutional kernels W_2 and fused with X_{2dwt} [as illustrated in (9) and (10)], resulting in the final output of the WSC module. Here, the choice of 3×3 convolutional kernels is attributed to their multifaceted advantages, including parameter count, computational efficiency, and training stability. The compact size of convolutional kernels results in a reduced parameter count, which mitigates the risk of overfitting during the training phase

$$X_{\text{2dwt}} = \text{IWT}_{\text{2d}}\left(\text{Conv}\left(W_1, \begin{bmatrix} X_{\text{LL}} & X_{\text{LH}} \\ X_{\text{HL}} & X_{\text{HH}} \end{bmatrix}\right)\right) \tag{8}$$

$$X_{\text{Conv2d}} = \text{Conv}\left(W_2, \text{PWConv}\left(X_{\text{input}}\right)\right) \tag{9}$$

$$X_{\text{spa}} = X_{\text{2dwt}} + X_{\text{Conv2d}}. \tag{10}$$

The WSC module leverages a 2-D wavelet transform to capture multiple frequency components within an image, recognizing key features such as shapes and contours. By incorporating 3×3 small convolutional kernels, the model's sensitivity to details is significantly enhanced. Furthermore, customized residual connection ensures the consistency and completeness of feature representation, thereby mitigating the risk of information loss. The integration of these mechanisms collectively contributes to improved classification accuracy, high computational efficiency, and robust model performance.

C. Spectral Wavelet Convolution

Unlike the spatial dimension, the spectral dimension carries a wealth of spectral information, with each pixel's response across multiple bands forming a unique spectral profile. To boost FE in spectral dimension of HSIs, we introduce the SWC module, which combines a 1-D wavelet transform with convolution, as depicted in Fig. 5.

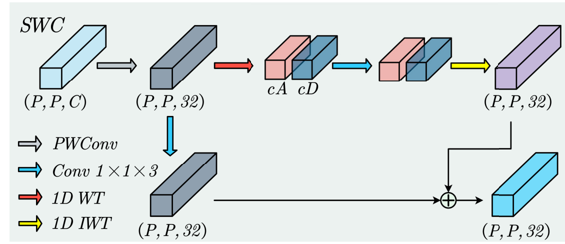


Fig. 5. Spectral FE process of SWC.

First, the input image patch $X_{\text{input}} \in \mathbb{R}^{P\times P\times C}$ undergoes a pointwise convolution along the spectral dimension, simplifying the spectral information without losing critical frequency details. It is important to highlight that the selection of 32 channels as the initial hyperparameter for the convolutional layers in both the SWC and WSC modules is determined through extensive experimentation. This choice is made to achieve an optimal balance between model complexity and performance. Subsequently, 1-D DWT is applied to decompose the dimensionality-reduced spectral feature maps into low-frequency component X_{cA} and high-frequency component X_{cD}. This process analyzes the spectral curve of each pixel along the spectral dimension, decomposing its frequency information into detailed components of various scales. The low-frequency component represents the overall trend of the spectral signature, while the high-frequency component captures fine variations within the spectrum. This frequency decomposition allows the model to focus on coarse-grained patterns and subtle variations in the spectral signatures. Following the 1-D wavelet transform, convolution operations are performed on the high- and low-frequency components, using a convolutional kernel $W_1 \in \mathbb{R}^{1\times 1\times 3}$ to extract features of different frequencies. Here, the utilization of $1 \times 1 \times 3$ convolutional kernels in the SWC module is attributed to their effectiveness in capturing interrelationships among adjacent frequency bands. This kernel configuration substantially diminishes the computational load and the volume of model parameters, thereby enhancing computational efficiency without compromising performance. Finally, the high- and low-frequency components are reintegrated back into the original spectral dimension through a 1-D inverse wavelet transform, as shown in the following equation:

$$X_{\text{1dwt}} = \text{IWT}_{\text{1d}}\left(\text{Conv}\left(W_1, \left[X_{\text{cA}}\ X_{\text{cD}}\right]\right)\right). \tag{11}$$

Concurrently, the original dimensionality-reduced feature maps undergo convolution through a layer with kernel $W_2 \in \mathbb{R}^{1\times 1\times 3}$ [as shown in (12)]. This convolution focuses on capturing local information within the spectral dimensions and extracting the relationships between adjacent bands for each pixel

$$X_{\text{Conv1d}} = \text{Conv}\left(W_2, \text{PWConv}\left(X_{\text{input}}\right)\right). \tag{12}$$

Similarly, the output of the SWC module employs a residual fusion, where the frequency-transformed feature maps X_{1dwt} are fused with X_{Conv} to produce the final output of the SWC module, as illustrated in the following equation:

$$X_{\text{spe}} = X_{\text{1dwt}} + X_{\text{Conv1d}}. \tag{13}$$

The SWC module effectively harnesses the strengths of the 1-D wavelet transform in spectral frequency decomposition, bolstering the model's sensitivity to refine spectral information. By integrating 1-D wavelet transform and convolution operations, the model achieves a more comprehensive capture of spectral features while maintaining relatively low computational complexity, thus substantially enhancing the classification performance. The frequency-domain features extracted via wavelet decomposition provide the subsequent Mamba block with rich multiscale contextual information. Meanwhile, the SSM mechanism within the Mamba block further enhances the model's selective perceptual capacity for these frequency-domain features through dynamic parameter adaptation. This adaptive process allows the model to dynamically allocate computational resources and adjust its sensitivity to specific frequency components based on the input data.

D. HyperMamba Block

The overall structure of the HyperMamba block is illustrated in Fig. 6, which consists of two parallel branches: spectral Mamba branch and spatial Mamba branch. The input feature maps are first normalized and then fed separately into two branches to capture spectral and spatial information, with further modeled through SSM. To enable efficient processing by the subsequent SSM mechanism, we systematically transform the extracted 3-D feature blocks into 2-D representations via 1-D scanning operations. This transformation is designed to preserve essential information from both frequency domain and spatial domain while facilitating the modeling of long-range dependencies by the SSM modules. Specifically, we employ 1-D scanning along predefined dimensions of each 3-D feature block to achieve dimensionality reduction, thereby generating 2-D feature blocks that retain critical contextual information. Unlike conventional spectral–spatial Mamba models that rely on fixed scanning strategies, the HyperMamba module introduces a more flexible approach. It generates unified token representations through a single scanning operation and dynamically performs axis permutations during subsequent processing steps. This adaptability allows the model to selectively scan different dimensions based on the input data characteristics.

In the spectral Mamba branch, the normalized feature sequences first pass through a linear layer LN_1 and a 1×1 convolution, preparing the features for selective scanning in subsequent modeling stages

$$X_1 = \mathrm{Conv}_{1\times 1}\left(\mathrm{LN}_1\left(\mathrm{Norm}\left(X_{\text{input}}\right)\right)\right). \tag{14}$$

The processed feature sequences X_1 are then passed to the SSM for further modeling. In the spectral Mamba branch, the SSM captures dynamic dependencies between features along spectral dimensions. Through leveraging its dynamic nature, the SSM enhances the model's ability to perceive global spectral information and improves its capacity to distinguish between different land-cover categories

$$Y_{\text{spe}} = \mathrm{SSM}(\mathbf{A}_1, \mathbf{B}_1, \mathbf{C}_1, X_1). \tag{15}$$

The processing flow of the spatial Mamba branch is akin to that of the spectral branch, with a crucial distinction: before entering the SSM, the input feature sequence undergoes a dimension transpose, transforming the data from spectral dimension to spatial dimension. This allows the SSM to model long-range dependencies between various spatial features. Through this process, the SSM effectively enhances the perception of global spatial information

$$X_2 = \mathrm{Conv}_{1\times 1}\left(\left(\mathrm{LN}_2\left(\mathrm{Norm}\left(X_{\text{input}}\right)\right)\right)^{\mathrm{T}}\right) \tag{16}$$

$$Y_{\text{spa}} = \left(\mathrm{SSM}(\mathbf{A}_2, \mathbf{B}_2, \mathbf{C}_2, X_2)\right)^{\mathrm{T}}. \tag{17}$$

Following the SSM processing in each branch, the activation branch modulates the SSM outputs using the SiLU activation function, as detailed in (18). This modulation enhances the nonlinear relationships within the feature representations. Subsequently, the activated spectral and spatial features are preliminarily fused through elementwise multiplication, as detailed in (19), ensuring the effective integration of multidimensional information. The fused representation is then combined with the original input features via a linear projection and residual connections, as detailed in (20). Specifically, the residual connections systematically integrate the original input features with the fused representations, ensuring both the preservation of primary information and the full exploitation of cross-dimensional feature interactions. This deep fusion mechanism not only enhances the model's capacity to comprehend and utilize complex informational relationships but also significantly improves classification performance. Through this architecture, the HyperMamba block achieves computationally efficient yet comprehensive joint modeling of spectral and spatial information

$$Z = \mathrm{SiLU}\left(\mathrm{LN}_3\left(\mathrm{Norm}\left(X_{\text{input}}\right)\right)\right) \tag{18}$$

$$\begin{aligned} Y'_{\text{spe}} &= Y_{\text{spe}} \odot Z \\ Y'_{\text{spa}} &= Y_{\text{spa}} \odot Z \end{aligned} \tag{19}$$

$$Y = \mathrm{LN}_4\left(Y'_{\text{spe}} + Y'_{\text{spa}}\right) + X_{\text{input}}. \tag{20}$$

It should be noted that four distinct linear layers (LN_1, LN_2, LN_3, and LN_4) are strategically deployed at different positions within the model architecture, each serving a specific purpose. LN_1 is responsible for conducting preliminary feature transformation on the raw input data. LN_2 is subsequently employed to further extract and enhance spectral features. LN_3 processes spatially permuted data following axis adjustment, ensuring effective handling of spatial dimensions. Finally, LN_4 integrates information from both spectral and spatial dimensions to generate unified feature representations.

The design of the HyperMamba block significantly enhances the capture of long-range dependencies while maintaining computational efficiency. By modeling feature sequences in parallel from both spectral and spatial dimensions, the HyperMamba block substantially improves the accuracy and robustness of HSIC.

IV. Experimental Results and Discussion

This section is organized as follows. Section IV-A describes four hyperspectral datasets used. Section IV-B presents ablation experiments on the wavelet-based FE block. Section IV-C analyzes the influence of the number of stacked layers in the

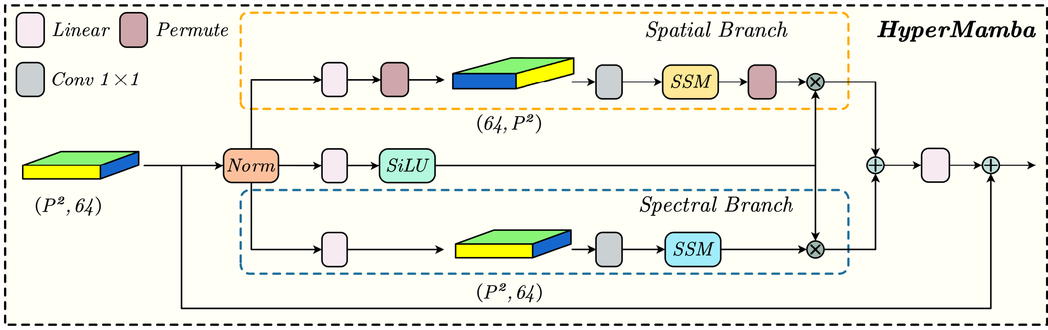


Fig. 6. HyperMamba block has two parallel branches: spatial branch and spectral branch. After normalization and a 1 × 1 convolution, the input is processed by both branches, which adjust dimensions and apply SSM for global information. Spatial and spectral features are fused using elementwise multiplication ("⊗") and addition ("⊕").

HyperMamba block. Section IV-D conducts branch ablation experiments on the HyperMamba block. Section IV-E analyzes the impact of different input patch sizes. Section IV-F compares the classification performance of our WD-SSMamba with several mainstream networks. Finally, Section IV-G provides the computational complexity analysis.

First, we introduce the parameters setting used in the experiments. During model training, the input image patches are normalized to a range of [0, 1]. In addition, we apply data augmentation techniques on the input image patches, including horizontal or vertical flips and random rotations of 90°, 180°, or 270°. Each original training sample can generate up to eightfold augmented data, where 20 training samples per class are expanded to 160 samples per class through data augmentation. We use the AdamW optimizer for parameters update, with a total of 250 training epochs. The learning rate is set to 0.001, the weight decay coefficient is 0.05, and the batch size is 128. The loss function used in the model is cross-entropy loss. To mitigate the impact of randomness and ensure the reliability of the results, we conduct each experiment ten times and record the average result.

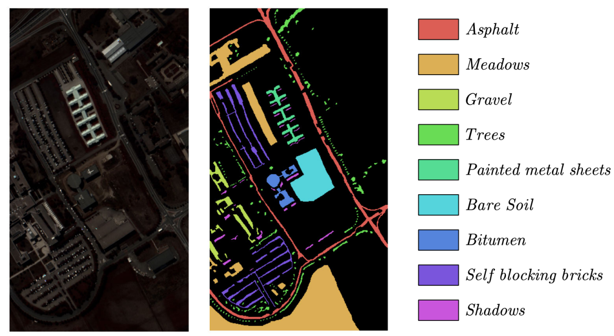
We use overall accuracy (OA), average accuracy (AA), and κ coefficient to evaluate classification performance. OA indicates the proportion of correctly predicted labels relative to the total number of labels, AA reflects the mean accuracy across all classes, and κ coefficient assesses the agreement between predicted labels and actual labels. It is worth noting that all our experiments are conducted on Ubuntu 22.04 equipped with a NVIDIA GeForce RTX 3090 GPU and 24 GB of memory, using Python 3.10 and PyTorch 2.1.0.

A. Hyperspectral Datasets

To validate the effectiveness of our proposed model, we adopt four publicly hyperspectral datasets: Pavia University, WHU-Hi-LongKou [46], [47], WHU-Hi-HongHu [46], [47], and Houston 2013. These datasets cover a variety of scenarios from urban to agricultural environments, providing a comprehensive assessment of the model's classification performance and generalization ability.

1) Pavia University: Pavia University dataset was acquired by the ROSIS sensor during a flight over Pavia, northern Italy. This dataset covers an urban environment with a spatial resolution of 1.3 m, containing 103 spectral bands and spatial dimensions of 610 × 340 pixels. It includes nine land-cover classes, such as asphalt, grass, gravel, etc. The division of training, verification, and testing sets is shown in Table I, with approximately 20 training samples per class.

2) WHU-Hi-LongKou: The WHU-Hi-LongKou dataset was collected from Longkou Town, Hubei, China, primarily for agricultural scene analysis. The data was acquired using an 8-mm focal length Headwall Nano-Hyperspec imaging sensor mounted on a DJI Matrice 600 Pro UAV platform at an altitude of 500 m. The dataset has a spatial resolution of approximately 0.463 m, a spectral range from 400 to 1000 nm, and includes 270 spectral bands [46], [47]. The image dimensions are 550 × 400 pixels, covering nine classes, including corn, cotton, sesame, rice, etc. The division of training, verification, and



(a) (b)

Fig. 7. (a) False color composition of Pavia University scene. (b) Ground-truth map containing nine mutually exclusive land-cover classes.

TABLE I
NUMBER OF TRAINING, VALIDATION, AND TESTING SAMPLES FOR EACH CLASS IN PAVIA UNIVERSITY DATASET

Class No.	Class Name	Train Num	Valid Num	Test Num
C1	Asphalt	22	63	6241
C2	Meadows	21	181	17965
C3	Gravel	16	18	1797
C4	Trees	21	29	2883
C5	Painted metal sheets	11	11	1102
C6	Bare Soil	21	46	4526
C7	Bitumen	15	10	971
C8	Self-Blocking Bricks	20	34	3330
C9	Shadows	9	8	787
	Total	156	400	39602

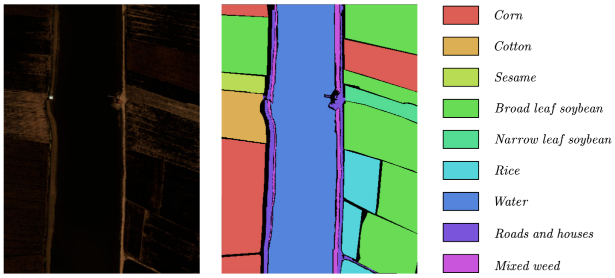


Fig. 8. (a) False color composition of the WHU-Hi-LongKou scene. (b) Ground-truth map containing nine mutually exclusive land-cover classes.

TABLE II
NUMBER OF TRAINING, VALIDATION, AND TESTING SAMPLES FOR EACH CLASS IN WHU-HI-LONGKOU DATASET

Class No.	Class Name	Train Num	Valid Num	Test Num
C1	Corn	20	172	34239
C2	Cotton	20	41	8233
C3	Sesame	20	15	2916
C4	Broad-leaf soybean	20	315	62797
C5	Narrow-leaf soybean	20	20	4031
C6	Rice	20	59	11695
C7	Water	20	335	66621
C8	Roads and houses	20	35	6989
C9	Mixed weed	20	36	5103
	Total	180	1028	202624

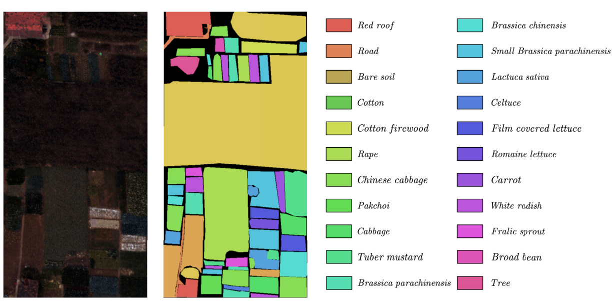


Fig. 9. (a) False color composition of the WHU-Hi-HongHu scene. (b) Ground-truth map containing 22 mutually exclusive land-cover classes.

testing sets is shown in Table II, with each class containing 20 training samples and a total of 202 624 testing samples.

3) WHU-Hi-HongHu: The WHU-Hi-HongHu dataset was acquired on November 20, 2017, in Honghu, Hubei, China, with a 17-mm focal length Headwall Nano-Hyperspec imaging sensor equipped on a DJI Matrice 600 Pro UAV platform. The dataset has a spatial resolution of approximately 0.043 m, a spectral range from 400 to 1000 nm, and includes 270 spectral bands [46], [47]. It comprises 22 land-cover classes, such as red roofs, bare soil, cotton, etc., with image dimensions of 940 $\times$ 475 pixels. The division of training, verification, and testing sets is shown in Table III, with each class containing 20 training samples and a total of 382 571 testing samples.

4) Houston 2013: The Houston 2013 dataset was collected using an airborne hyperspectral imaging sensor over Houston, TX, USA. This dataset has a spatial resolution of 2.5 m, includes 349 $\times$ 1905 pixels, a spectral range from 380 to 1050 nm, and contains 144 spectral bands. It provides a rigorous testing platform for HSIC in complex urban environments. The dataset includes 15 land-cover classes, such as healthy grass, buildings, roads, railways, etc. The division of training, verification, and testing sets is shown in Table IV.

TABLE III
NUMBER OF TRAINING, VALIDATION, AND TESTING SAMPLES FOR EACH CLASS IN WHU-HI-HONGHU DATASET

Class No.	Class Name	Train Num	Valid Num	Test Num
C1	Red roof	20	70	13871
C2	Road	20	17	3395
C3	Bare soil	20	109	21612
C4	Cotton	20	816	162369
C5	Cotton firewood	20	31	6087
C6	Rape	20	222	44235
C7	Chinese cabbage	20	120	23883
C8	Pakchoi	20	20	3934
C9	Cabbage	20	54	10665
C10	Tuber mustard	20	61	12233
C11	Brassica parachinensis	20	54	10861
C12	Brassica chinensis	20	44	8810
C13	Small Brassica chinensis	20	112	22295
C14	Lactuca sativa	20	36	7220
C15	Celtuce	20	4	898
C16	Film covered lettuce	20	36	7126
C17	Romaine lettuce	20	14	2896
C18	Carrot	20	16	3101
C19	White radish	20	43	8569
C20	Garlic sprout	20	17	3369
C21	Broad bean	20	6	1222
C22	Tree	20	20	3920
	Total	440	1922	382571

TABLE IV
NUMBER OF TRAINING, VALIDATION, AND TESTING SAMPLES FOR EACH CLASS IN HOUSTON 2013 DATASET

Class No.	Class Name	Train Num	Valid Num	Test Num
C1	Healthy grass	20	53	1000
C2	Stressed grass	19	53	1011
C3	Synthetic grass	19	25	480
C4	Trees	19	53	1003
C5	Soil	19	53	1003
C6	Water	18	7	136
C7	Residential	20	53	1019
C8	Commercial	19	52	1001
C9	Road	19	53	1006
C10	Highway	19	52	984
C11	Railway	18	53	1001
C12	Parking Lot 1	19	52	989
C13	Parking Lot 2	18	14	271
C14	Tennis Court	18	12	235
C15	Running Track	19	24	449
	Total	283	609	11588

To address the significant concern of spatial overlap between the training and testing sets, we selected all training samples from the officially provided training sets of three benchmark datasets: Houston, WHULK, and WHUHH. These datasets have predefined nonoverlapping splits for training and testing, ensuring spatial separation. For Pavia University dataset, we focused on evaluating model performance with limited training samples. Consequently, we restricted the number of training samples for each class to less than 20, comprising only 0.04% of the total samples. This extremely small proportion

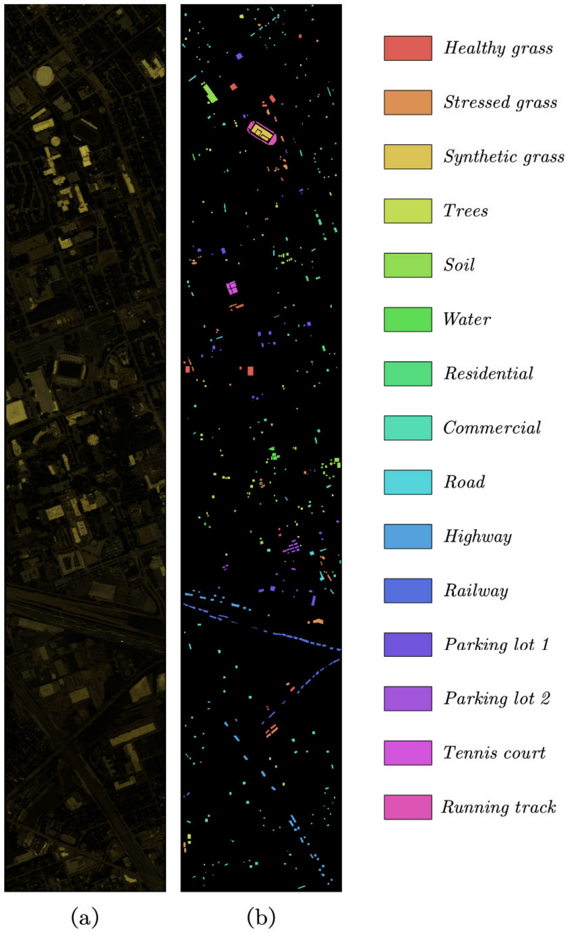


Fig. 10. (a) False color composition of the Houston 2013 scene. (b) Ground-truth map containing 15 mutually exclusive land-cover classes.

of training data, coupled with the relatively small patch size, effectively minimized the risk of overlap between the training and testing sets.

B. Ablation Study on the FE Block

In this experiment, we conducted ablation studies on the WSC and SWC branches of the FE block to assess their role and effectiveness in HSIC tasks. Table V illustrates the impact of the WSC and SWC blocks on model performance.

First, when both WSC and SWC modules were completely removed, the model's classification performance significantly declined. These findings indicate that, without these two components, the model fails to fully leverage the abundant frequency information present in HSIs, leading to inferior classification outcomes. Subsequently, we conducted experiments using only SWC or only WSC. The results revealed that WSC obtained better OA than SWC, suggesting that spatial frequency features possess superior discriminative ability compared with spectral frequency features. However, the classification performance of the WSC alone did not reach optimal levels. The outcomes from both scenarios imply that, while individual spatial or spectral frequency features contribute to the model's performance, they do not fully unlock the model's potential on their own. When both WSC and SWC modules were used together, the model achieved the best results across all datasets. The WSC module extracts multiscale spatial-frequency features through 2-D wavelet decomposition. In parallel, the SWC module employs 1-D wavelet decomposition to separate low-frequency trends and high-frequency details in spectral curves. This dual-module design achieves complementary FE through coordinated operation. This outcome underscores the synergistic effect of these two modules, as they work together to capture and integrate the multidimensional information within HSIs.

TABLE V
OA With and Without WSC and SWC Modules Across Different Datasets

Feature Extraction		Overall Accuracy (OA)			
WSC	SWC	PaviaU	WHU-LK	WHU-HH	Houston
×	×	90.51	97.07	89.06	85.71
✓	×	92.07	97.82	89.51	86.42
×	✓	91.17	97.35	89.32	86.11
✓	✓	**92.48**	**97.91**	**89.98**	**87.00**

C. Analysis on the Number of HyperMamba Blocks

In this experiment, we focused on investigating the impact of the HyperMamba block on the model's performance. The experimental results are displayed in Table VI, which presents the OA, the number of Params, and FLOPs for various datasets under varying numbers of stacked HyperMamba blocks.

As shown in Table VI, when the number of HyperMamba blocks is 0, indicating that the HyperMamba block is not used, the classification accuracy for all datasets is notably lower. For instance, the OA for the Houston 2013 dataset is 86.07% $\pm$ 1.75%, and for Pavia University dataset, it is merely 89.77% $\pm$ 3.04%. This demonstrates that models without the HyperMamba block possess limited classification capabilities when processing HSIs, as they fail to adequately capture global spectral–spatial features.

With the inclusion of one HyperMamba block, there is a notable improvement in classification accuracy across all datasets. Specifically, the performance with one HyperMamba block clearly outperforms the scenario without any blocks. For instance, the OA for the Houston 2013 dataset rises to 87.00% $\pm$ 1.90%, and for Pavia University dataset, it reaches 92.48% $\pm$ 1.03%. In WHU-Hi-LongKou and WHU-Hi-HongHu datasets, the inherently rich spectral and spatial characteristics, coupled with high information density, enable the existing wavelet FE block to generate high-quality feature representations. Consequently, the incorporation of an additional HyperMamba block yields relatively marginal performance improvements in these two scenarios. This outcome demonstrates that the HyperMamba block effectively boosts classification performance while keeping relatively low computational costs, as evidenced by the parameter count of 33.97k for Pavia University dataset.

However, when the stacking depth of HyperMamba blocks exceeds one, despite a substantial increase in both computational and parameter costs, the classification accuracy actually diminishes. Specifically, with two HyperMamba blocks stacked, the OA for the Houston 2013 dataset drops to

85.48% ± 2.11%, and for Pavia University dataset, it falls to 91.93% ± 2.14%. This phenomenon suggests that an excessive number of HyperMamba blocks does not further enhance model performance; instead, the heightened computational complexity and potential overfitting result in a decline in classification effectiveness. For example, while the parameter count for Pavia University dataset increases from 33.97k to 54.89k, the classification accuracy decreases correspondingly.

In summary, using one HyperMamba block strikes an optimal balance between classification accuracy and computational overhead. Conversely, stacking more blocks increases computational costs without yielding the expected performance improvements, and in some cases, it even leads to a drop in accuracy. Therefore, the configuration with one HyperMamba block demonstrates the best performance in this experiment, as it significantly enhances classification performance while avoiding unnecessary computational burdens.

D. Ablation Study on the Branches of the HyperMamba

In this experiment, we systematically assessed the individual contributions of the spectral Mamba branch and the spatial Mamba branch to the model's performance. As evident from Table VII, retaining only the spatial Mamba branch resulted in a decline in the model's classification accuracy. While the spatial Mamba branch effectively captured spatial information, the absence of spectral processing hindered the model's ability to fully identify critical features, leading to a drop in performance across multiple datasets. Conversely, retaining only the spectral Mamba branch also resulted in a decline in performance due to the loss of spatial feature modeling, confirming that neither spatial nor spectral information alone is sufficient for optimal classification. Finally, when both branches were used together, the highest classification accuracies were achieved across all datasets. However, compared with the two datasets of PaviaU and Houston, the performance improvement on the other two high-spatial resolution datasets (WHU-LK and WHU-HH) is relatively limited. The reason is that the enhanced spatial redundancy inherent in high-spatial resolution images (where individual pixels contain rich ground object information with minimal variations among adjacent pixels) necessitates special processing of local details. Our wavelet convolution addresses this through its exceptional localization capability in time–frequency analysis, achieving effective multiscale spatial–spectral FE, thereby compressing redundant information while preserving critical local features (e.g., textures and edges). This is also the fundamental motivation for our wavelet integration. In contrast, although Mamba is better at handling long-sequence modeling, high-spatial resolution image classification exhibits strong local dominance where single pixels/small regions often contain complete ground object information. In such cases, the Mamba architecture's global modeling advantages might be counterbalanced by noise amplification from redundant pixel relationships. Furthermore, the inherent high similarity between adjacent pixels challenges Mamba's selective scanning mechanism in effectively filtering spatial redundancies, resulting in the improvement brought by global modeling is limited, e.g., repeatedly modeling the relationship of similar pixels. Therefore, to address these fundamental challenges, we design a lightweight hybrid architecture synergizing wavelet decomposition with enhanced Mamba operations. This design philosophy strategically combines localized feature refinement (through wavelet-based redundancy compression) with optimized global modeling (via sequence-length-reduced Mamba processing).

To further validate the superiority of the Mamba architecture over conventional Transformer models, we conducted an ablation study in which the entire HyperMamba block was replaced with a standard Transformer Encoder (equipped with four-head attention and identical hidden dimensions). The experimental results demonstrate that our HyperMamba module consistently outperforms the Transformer encoder across all tested datasets. This result once again validates the design of the HyperMamba block, demonstrating that the combined use of the spectral and spatial Mamba branches is essential for capturing multidimensional information in HSIs. Consequently, this integration significantly enhances the model's overall performance.

E. Analysis on Different Patch Sizes

In HSIC tasks, image patches are typically used as input to capture spectral information while fully harnessing spatial context. However, the selection of patch size has a notable impact on model performance. Patches that are too small may not effectively exploit spatial information, whereas excessively large patches can elevate computational burdens and introduce redundant information, ultimately detrimental to classification outcomes.

In our experiment, we evaluated OA, Params, and FLOPs for various patch sizes across four datasets. It is worth noting that WD-SSMamba does not require the patch size to be constrained as either odd or even. In both cases, the label of each patch is determined by the label of its geometric center pixel. Specifically, when the patch size is of even dimensions (P, P, C) with P being even, the corresponding center pixel is located at $((P/2), (P/2))$. However, in our comparative experiments, some of the methods we compared with, such as GSC-ViT [48], also adopted even patch sizes. Therefore, to ensure the fairness of the model comparison, we set all patch sizes to be even in our experiments.

As shown in Fig. 11, the visualized results unmistakably showcase that alterations in patch size exert a significant influence on model performance. Notably, an increase in patch size from 6×6 to 8×8 yields a substantial improvement in classification accuracy. For instance, in Pavia University dataset, the OA rises from 89.68% to 92.48% with this increase in patch size. Similarly, in the Houston 2013 dataset, the OA rises from 84.31% to 87.00%. This increment comes with a moderate computational overhead, as the parameter count and FLOPs remain within manageable limits of 33.97k and 2.5M, respectively.

However, further increasing the patch size yields diminishing gains in accuracy, and in some datasets, it even leads to a decline in accuracy. For example, in the WHU-Hi-LongKou dataset, increasing the patch size from 8×8 to 10×10 causes

TABLE VI

OA (%), PARAMETERS (K), AND FLOPS (M) WITH DIFFERENT NUMBERS OF STACKED HYPERMAMBA BLOCKS ACROSS DIFFERENT DATASETS

Blocks	Pavia University			WHU-Hi-LongKou			WHU-Hi-HongHu			Houston 2013		
	OA(%)	Params	FLOPs	OA(%)	Params	FLOPs	OA(%)	Params	FLOPs	OA(%)	Params	FLOPs
0	89.77±3.04	13.04	0.75	97.64±0.73	23.73	1.43	89.18±0.48	24.57	1.43	86.07±1.75	16.05	0.92
1	**92.48±1.03**	33.97	2.33	**97.91±0.65**	44.65	3.02	**89.98±0.66**	45.50	3.02	**87.00±1.90**	36.98	2.50
2	91.93±2.14	54.89	3.92	97.67±0.66	65.58	4.61	89.83±0.43	66.43	4.61	85.48±2.11	57.91	4.09
3	90.59±2.20	75.82	5.51	97.70±0.64	86.51	6.20	89.44±0.75	87.35	6.20	85.33±2.06	78.84	5.68

TABLE VII

OA WITH AND WITHOUT SPECTRAL AND SPATIAL BRANCH ACROSS DATASETS

Mamba Branch		Overall Accuracy (OA)			
Spatial	Spectral	PaviaU	WHU-LK	WHU-HH	Houston
✓	×	90.90	97.74	89.66	86.16
×	✓	91.68	97.66	89.85	86.08
✓	✓	**92.48**	**97.91**	**89.98**	**87.00**
Transformer Encoder		89.56	97.68	89.30	85.57

a slight decrement in OA, from 97.91% to 97.72%. Similarly, in the WHU-Hi-HongHu dataset, while accuracy does improve with larger patch sizes, the increase is far less substantial compared with the leap from 6 × 6 to 8 × 8. Concurrently, as the patch size increases, both the parameter count and FLOPs escalate drastically. For instance, in Pavia University dataset, increasing the patch size from 8 × 8 to 14 × 14 causes a sharp surge in FLOPs, from 2.33M to 8.81M, despite an improvement in accuracy by less than 1%.

In summary, an 8 × 8 patch size achieves the optimal balance across all datasets. It notably improves classification accuracy while avoiding the steep rise in computational costs and potential redundant information that associated with larger patch sizes. Consequently, we ultimately select an 8 × 8 patch size as the input configuration for our model to guarantee optimal performance and efficiency.

F. Classification Comparison With State-of-the-Art Models

In this study, we conducted a comprehensive comparison of our proposed WD-SSMamba network with several state-of-the-art models, including three CNN-based models, four Transformer-based models, and one Mamba-based model. The experiments were performed on four HSI datasets, with approximately 20 training samples per class for each dataset. For a fair comparison, the patch size for all compared models and datasets was set to 8 × 8.

1) *3DCNN [49]:* This model consists of three 3-D convolutional layers, two 3-D pooling layers, one dropout layer, and one fully connected layer. The first two 3-D convolutional layers are followed by two 3-D pooling layers, and finally, a fully connected layer is used for classification output.
2) *SPRN [50]:* This model primarily includes a padding layer, an optional spatial attention layer, two convolutional layers, two residual blocks, one fully connected layer, and one linear classification layer.
3) *CLOLN [51]:* The model contains two spectral–spatial modules (COS2M) and one 2-D convolutional layer. Each COS2M module combines dual single-channel 3-D convolution and depthwise convolution for efficient spectral–spatial extraction.
4) *SpectralFormer [12]:* This model includes a layer for processing spectral information from adjacent bands, a patch embedding layer, a Transformer layer, and an MLP layer for classification.
5) *SSFTT [13]:* This model includes a 3-D convolutional layer, a 2-D convolutional layer, a Transformer layer with positional encoding, a dropout layer, and a linear classification layer.
6) *GAHT [52]:* This model consists of three convolutional layers, three Transformer layers, a global average pooling layer, and a linear classification layer.
7) *GSC-ViT [48]:* The model includes a spectral embedding layer, Transformer layers, and an MLP classification layer. The spectral embedding extracts adjacent band information, the Transformer captures local spectral sequences, and the MLP performs the final classification.
8) *MiM [53]:* This model integrates a convolutional module, a Mamba-cross-scan module, and a fully connected layer. The convolutional module first extracts features, which are then enhanced with spatial information through the cross-scan mechanism, and finally classified through the fully connected layer.

Tables VIII–XI present the OA, AA, κ, and accuracy per class for various models on four HSI datasets, highlighting the best results in bold. Evidently, our proposed model outperformed all others, achieving the highest OA, AA, and κ across all datasets.

On Pavia University dataset, our method achieved an OA of 92.48%, leading in AA (89.96%) and κ (89.87) with minimal standard deviation, showcasing its stability and reliability in urban scenarios.

For the WHU-Hi-LongKou dataset, our model excelled in classifying cotton (98.37%) and broad-leaf soybean (95.60%), achieving an OA of 97.91%, and ranking first in AA (97.07%) and κ (97.27%), and SPRN achieved the second-best accuracy. SPRN demonstrates strength in local feature modeling through residual learning and spatial–spectral attention; its convolutional backbone inherently limits long-range dependency capture. In contrast, our WD-SSMamba provides a HyperMamba module to establish global spectral–spatial correlations, which not only enhances the extraction of complex data patterns but also maintains high computational efficiency. Therefore, WD-SSMamba enables more efficient extraction of discriminative patterns from high-dimensional spectral data, thus providing better classification performance under certain circumstances.

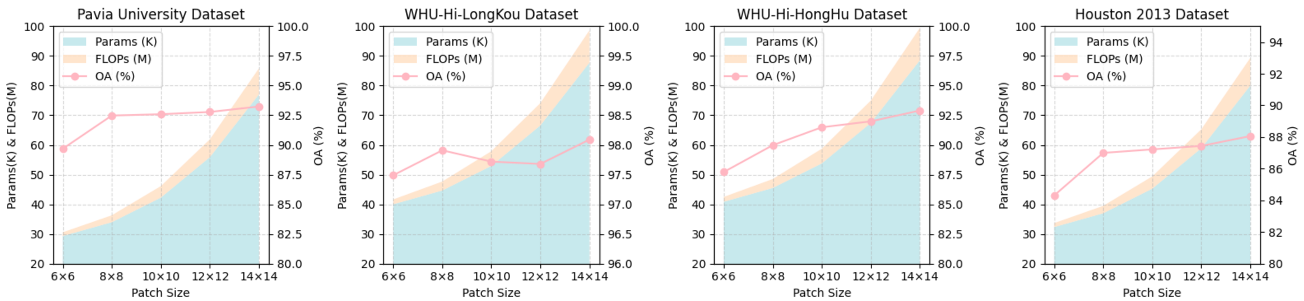


Fig. 11. OA, Params, and FLOPs by using different patch sizes on four datasets.

TABLE VIII

OAs (%), AAs (%), AND $\kappa \times 100$ STATISTIC FOR PAVIA UNIVERSITY DATASET

Class	CNN based Networks			Transformer based Networks				Mamba based Networks	
	3DCNN	SPRN	CLOLN	Spectralformer	SSFTT	GAHT	GSC-ViT	MiM	Ours
1	69.88±6.06	88.55±6.66	85.60±6.91	64.07±8.55	80.05±6.57	90.08±2.73	**91.04±2.37**	87.09±3.28	88.85±4.38
2	84.36±10.51	89.27±9.00	87.81±10.56	71.69±11.77	**96.37±1.77**	95.04±1.97	92.53±3.90	84.15±11.39	96.32±2.77
3	36.52±18.49	64.79±27.24	59.41±18.52	59.12±10.77	63.19±10.77	61.46±9.79	**72.10±14.59**	65.79±16.11	63.11±13.15
4	92.19±4.00	88.87±3.75	94.48±1.62	85.66±5.94	93.28±7.48	**95.32±1.55**	94.47±3.30	89.63±3.76	94.19±3.75
5	98.53±1.80	99.74±0.23	99.46±0.70	99.85±0.20	98.94±1.59	99.67±0.51	99.39±0.81	**99.99±0.03**	99.86±0.23
6	44.21±12.92	85.39±5.41	91.76±5.98	57.59±12.46	74.23±11.98	82.45±11.11	77.10±14.23	83.82±12.91	**89.50±7.24**
7	67.98±8.76	71.00±32.89	76.58±13.74	51.41±7.37	71.79±12.71	77.02±7.19	84.10±12.86	86.68±5.96	**87.99±10.70**
8	92.08±3.53	91.90±6.46	94.35±3.35	76.83±9.11	90.84±3.36	92.37±2.61	93.25±3.65	**95.89±1.90**	95.43±1.95
9	96.52±3.13	96.01±2.66	**97.88±2.26**	88.72±5.32	93.96±5.49	95.43±2.25	96.19±3.20	91.26±4.37	94.38±6.26
OA(%)	76.77±4.37	87.77±3.96	87.91±4.51	70.38±4.97	88.50±1.53	90.79±1.33	89.86±2.22	85.77±4.19	**92.48±1.03**
AA(%)	75.81±2.42	86.17±6.97	87.48±2.05	72.77±1.72	84.74±2.21	87.65±1.37	88.91±3.20	87.14±2.09	**89.96±2.09**
$\kappa \times 100$	68.98±4.86	83.76±4.89	84.14±5.44	61.72±5.19	84.38±2.11	87.56±1.83	86.36±2.97	81.36±4.98	**89.87±1.35**

TABLE IX

OAs (%), AAs (%), AND $\kappa \times 100$ STATISTIC FOR THE WHU-HI-LONGKOU DATASET

Class No.	CNN based Networks			Transformer based Networks				Mamba based Networks	
	3DCNN	SPRN	CLOLN	Spectralformer	SSFTT	GAHT	GSC-ViT	MiM	Ours
1	97.44±1.09	**99.83±0.19**	99.79±0.34	99.42±0.42	99.16±0.81	99.52±0.59	99.51±0.35	99.50±0.77	99.79±0.44
2	78.20±6.04	97.02±1.90	96.34±3.37	86.55±6.59	97.92±1.16	98.12±0.85	98.20±1.57	94.31±5.28	**98.37±0.80**
3	91.39±5.35	**99.20±0.58**	96.73±3.53	95.83±4.60	96.76±4.83	99.17±0.59	94.68±3.73	95.00±11.73	98.54±1.61
4	81.53±3.59	95.59±2.10	92.85±2.81	85.98±3.32	92.73±2.06	92.46±2.90	95.12±2.49	90.02±1.68	**95.60±2.07**
5	83.64±8.39	95.28±2.97	95.03±2.54	77.73±6.79	92.67±4.50	**97.86±0.66**	95.70±2.28	96.50±2.70	95.41±3.29
6	96.93±1.67	**99.02±0.54**	98.52±1.19	89.15±3.89	98.19±1.41	97.81±1.38	97.93±1.65	97.70±2.29	98.57±1.44
7	**99.97±0.03**	99.73±0.27	99.79±0.23	99.86±0.17	99.82±0.19	99.40±0.47	99.73±0.32	99.89±0.13	99.80±0.22
8	87.15±6.34	94.03±3.83	93.12±2.95	81.16±4.05	93.76±4.19	**95.08±3.01**	91.99±5.33	93.42±2.02	94.91±3.12
9	86.79±8.31	88.86±9.05	92.51±5.20	71.66±7.78	89.60±10.71	92.16±6.33	86.81±10.72	87.94±7.06	92.68±3.58
OA(%)	91.55±1.13	97.74±0.56	96.87±0.83	92.47±1.10	96.69±0.49	96.76±0.91	97.35±0.78	95.75±0.69	**97.91±0.65**
AA(%)	89.23±1.53	96.51±1.35	96.08±0.69	87.48±1.64	95.62±1.27	96.84±0.73	95.52±1.51	94.92±2.25	**97.07±0.46**
$\kappa \times 100$	89.09±1.41	97.05±0.73	95.92±1.07	90.24±1.39	95.67±0.64	95.78±1.18	96.54±1.01	94.46±0.90	**97.27±0.84**

On the WHU-Hi-HongHu dataset, our model maintained top performance with an OA of 89.98%, AA of 86.87%, and κ of 87.39, exhibiting stability in large-scale multi-class datasets, and GAHT achieved the second-best accuracy. GAHT fuses local-global features through grouped pixel embedding (GPE) and hierarchical transformer structures; its fixed-window attention mechanism struggles to adapt to complex spatial distributions. WD-SSMamba, however, adopts a dynamic scanning strategy to adaptively select critical bands and spatial locations, combined with multiscale frequency-domain filtering capabilities via wavelet decomposition. This frequency-domain and spatial-domain collaborative mechanism make the model more robust in complex scenes and capable of uncovering deep correlations even with imbalanced sample distribution datasets.

For the Houston 2013 dataset, MiM achieved the second-best accuracy, as MiM captures local contextual information through a multidirectional scanning mechanism but lacks explicit frequency-domain modeling capability. While our WD-SSMamba model showed clear advantages, particularly in commercial (8), road (9), and railway (11) categories, achieving an OA of 87.00%, AA of 88.48%, and κ of 85.88. It is because our WD-SSMamba separates spectral and spatial features into low-frequency and high-frequency components using wavelet decomposition, making full use of frequency-domain information. The purely spatial scanning approach of MiM is prone to mixed-pixel interference, limiting performance gains. In addition, MiM's need to independently process features for each scanning path through the multidirectional scanning (MDS) mechanism increases

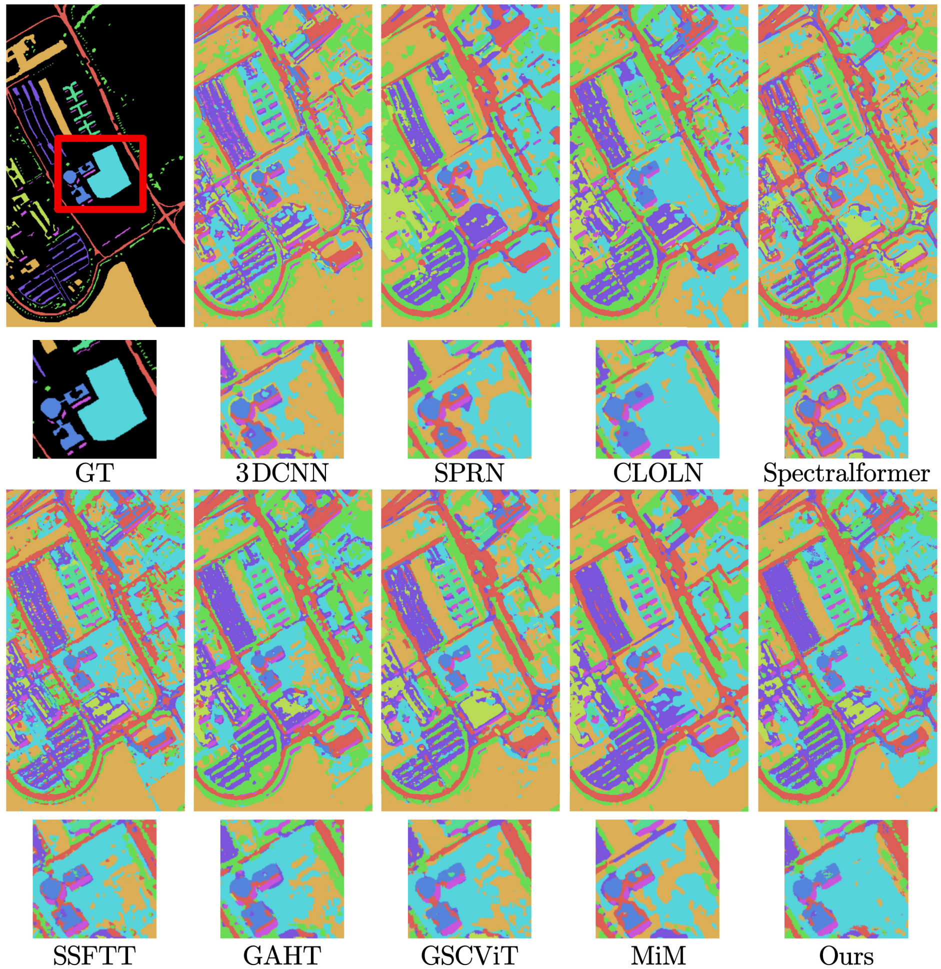


Fig. 12. Classification maps obtained by different models for Pavia University dataset. These maps correspond to a single experiment in Table VIII.

TABLE X

OAs (%), AAs (%), AND $\kappa \times 100$ STATISTIC FOR THE WHU-HI-HONGHU DATASET

Class	CNN based Networks			Transformer based Networks				Mamba based Networks	
	3DCNN	SPRN	CLOLN	Spectralformer	SSFTT	GAHT	GSC-ViT	MiM	Ours
1	78.00±7.38	91.93±6.74	90.67±6.24	91.39±6.03	90.20±5.76	92.52±5.64	90.72±6.42	**93.93±3.38**	90.74±5.50
2	71.97±15.09	85.86±4.90	82.06±9.03	78.25±7.62	78.39±11.17	87.70±8.57	88.12±7.78	90.74±2.96	**91.78±3.43**
3	80.81±7.23	83.55±3.03	82.61±4.49	79.62±4.04	86.12±4.81	**86.47±2.81**	86.32±4.36	82.75±3.41	84.67±3.50
4	89.10±4.05	95.14±1.06	94.65±1.77	93.50±3.06	93.21±1.85	96.17±1.46	95.76±1.11	94.12±2.14	**96.98±1.52**
5	57.47±11.95	**86.89±5.45**	78.69±9.99	80.13±5.42	75.08±11.25	83.93±9.37	83.46±9.09	86.28±5.18	79.16±8.36
6	85.49±2.83	90.85±3.43	89.26±3.17	85.25±4.03	89.71±3.03	88.65±3.60	89.29±3.57	90.57±2.14	**91.14±2.66**
7	59.22±3.90	74.28±3.52	70.32±4.32	66.84±5.35	66.25±3.62	76.34±5.47	75.47±4.97	**76.98±6.24**	75.69±7.05
8	27.25±7.24	58.91±7.92	57.05±7.92	43.81±4.59	49.52±7.22	63.76±9.89	62.13±7.62	69.29±3.04	**69.71±9.13**
9	89.02±3.56	95.06±1.17	94.62±1.62	95.20±1.64	94.03±2.01	**95.70±1.36**	95.12±1.63	95.06±1.74	95.19±1.49
10	57.39±9.15	76.69±4.38	74.79±7.75	54.29±4.65	76.73±6.20	82.21±3.89	80.97±9.22	79.51±4.96	**83.11±5.85**
11	39.88±7.37	**79.79±4.66**	68.65±7.67	60.98±7.43	65.99±5.44	74.13±6.29	75.35±6.22	77.60±7.11	77.26±7.65
12	55.62±12.96	**74.31±6.45**	68.38±10.74	57.31±5.04	66.25±8.45	72.92±6.90	70.96±6.73	71.85±5.03	72.77±6.35
13	55.69±8.20	67.02±7.22	68.40±4.98	62.28±6.00	71.06±7.21	**73.30±5.18**	72.53±5.82	72.11±3.84	72.45±8.00
14	66.74±6.70	89.94±3.15	87.68±3.79	76.21±5.67	84.20±7.00	88.84±4.06	86.05±7.27	**91.68±2.48**	89.31±5.71
15	85.27±4.54	97.38±2.08	95.79±2.97	91.77±3.10	94.99±2.18	97.98±1.02	96.08±1.56	96.14±3.03	**98.30±1.11**
16	77.53±8.29	91.59±2.60	90.03±5.40	82.86±6.83	89.05±4.96	91.44±2.52	92.86±4.49	92.10±2.71	**93.95±3.00**
17	75.44±9.78	95.11±3.60	88.94±20.64	83.73±8.06	94.46±4.04	96.21±4.38	94.59±6.21	**98.37±0.64**	95.80±3.17
18	77.57±4.67	91.54±5.10	89.97±7.57	83.71±5.77	86.95±7.20	92.85±3.87	86.47±9.48	89.74±4.14	**93.12±2.91**
19	81.28±3.43	90.72±3.19	88.56±5.69	81.58±3.43	87.81±4.60	91.99±2.54	89.77±3.27	87.57±3.71	**92.72±4.14**
20	87.83±2.41	96.06±1.63	94.20±2.54	91.79±1.73	90.34±4.06	94.52±1.36	94.77±3.81	93.97±1.45	**96.52±2.12**
21	57.89±23.33	**86.05±11.40**	69.29±26.14	73.79±9.15	67.36±25.66	77.48±15.89	72.55±20.64	85.71±8.90	75.07±27.06
22	80.01±5.39	93.52±3.08	90.69±3.58	86.46±6.56	85.95±9.80	92.67±4.03	93.88±2.90	91.34±4.95	**95.59±3.05**
OA(%)	78.30±1.93	88.51±0.71	86.86±0.91	83.28±0.97	85.97±1.01	89.37±0.72	88.87±0.81	88.59±0.77	**89.98±0.66**
AA(%)	69.84±2.44	86.01±0.64	82.51±2.97	77.31±1.31	81.08±2.53	86.26±0.99	85.15±1.18	86.70±0.72	**86.87±1.75**
$\kappa \times 100$	73.20±2.18	85.62±0.86	83.58±1.11	79.17±1.10	82.49±1.22	86.67±0.86	86.04±0.99	85.75±0.90	**87.39±0.82**

computational complexity with the number of directions. In contrast, WD-SSMamba's HyperMamba module employs a single bidirectional scan, significantly reducing computational complexity.

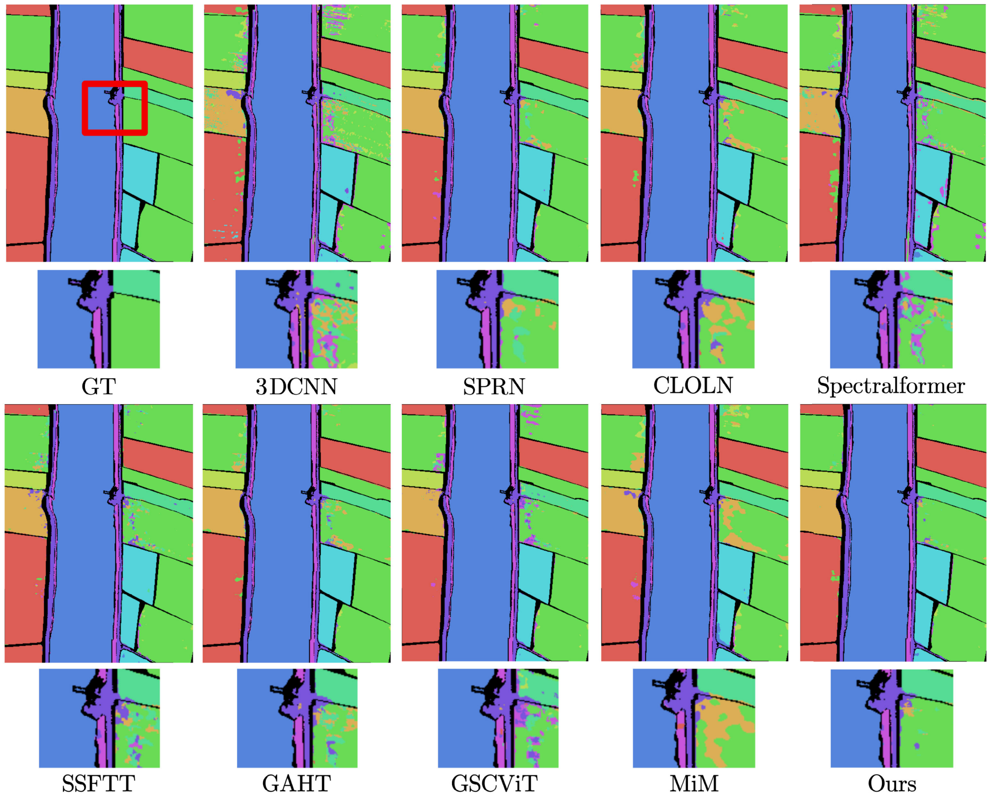


Fig. 13. Classification maps obtained by different models for the WHU-Hi-LongKou dataset. These maps correspond to a single experiment in Table IX.

TABLE XI

OAs (%), AAs (%), AND $\kappa \times 100$ STATISTIC FOR THE HOUSTON 2013 DATASET

Class	CNN based Networks			Transformer based Networks				Mamba based Networks	
	3DCNN	SPRN	CLOLN	Spectralformer	SSFTT	GAHT	GSC-ViT	MiM	Ours
1	81.99±1.29	**82.78±0.30**	82.53±0.58	80.93±1.23	82.25±0.48	82.00±0.66	81.19±2.01	82.08±0.97	82.40±0.53
2	78.81±6.05	**99.52±0.51**	98.28±2.22	89.27±7.71	97.26±1.81	98.62±1.37	98.45±1.45	97.86±1.45	98.54±1.27
3	91.37±1.91	**98.69±0.76**	98.02±2.09	90.71±3.62	96.98±3.51	98.52±1.28	98.35±1.94	98.38±1.07	97.48±2.01
4	94.05±3.32	95.91±1.62	97.43±3.10	86.16±2.60	**98.91±0.85**	97.56±1.69	96.41±4.57	97.94±1.22	97.55±2.13
5	92.44±6.33	98.84±2.57	**99.46±1.14**	95.29±4.37	96.36±5.64	99.06±2.33	98.19±3.76	99.34±1.39	98.62±2.46
6	78.46±4.44	92.43±5.63	89.93±4.89	77.79±4.36	91.25±7.27	90.66±5.49	**95.51±4.89**	91.47±4.77	87.65±7.81
7	**90.76±4.18**	88.22±4.84	83.13±7.05	74.97±5.79	84.62±8.55	89.67±8.17	88.16±4.79	87.85±3.27	88.48±4.16
8	60.41±5.47	67.04±9.42	62.32±11.50	60.17±6.09	65.69±10.72	74.17±5.65	71.02±7.63	77.09±5.59	**77.26±6.72**
9	77.43±3.61	78.25±3.28	77.96±4.91	62.80±5.67	78.62±4.53	79.47±4.14	79.11±3.42	77.50±4.09	**80.63±4.45**
10	44.43±12.85	62.16±12.84	60.41±19.50	49.15±7.97	60.22±18.11	59.08±18.79	64.34±18.02	**72.30±14.97**	65.08±22.74
11	77.11±4.19	**87.09±6.06**	77.43±11.59	63.30±4.44	72.66±14.44	77.11±12.25	81.28±9.81	81.83±9.21	**87.09±8.32**
12	52.39±7.93	77.54±7.69	70.16±10.13	67.20±6.80	71.59±9.64	79.75±7.75	74.11±8.14	79.54±7.10	**79.85±5.71**
13	84.43±6.98	75.20±6.41	87.01±4.72	49.00±10.88	90.52±4.14	**91.22±1.34**	85.09±3.17	83.84±5.27	89.48±3.21
14	88.26±5.51	**99.28±0.97**	96.26±4.09	76.51±7.25	94.43±4.70	98.13±2.35	97.23±4.01	98.64±0.85	97.70±2.73
15	95.50±1.87	**99.73±0.80**	98.82±1.42	91.29±6.63	98.15±1.32	**99.73±0.54**	99.49±1.07	99.11±0.54	99.42±0.75
OA(%)	77.09±1.99	85.26±1.90	82.94±2.51	74.00±1.73	82.85±1.16	85.51±2.08	85.02±1.06	86.76±1.58	**87.00±1.90**
AA(%)	79.19±1.76	86.85±1.64	85.28±1.90	74.30±1.96	85.30±1.28	87.65±1.58	87.20±1.00	88.32±1.53	**88.48±1.75**
$\kappa \times 100$	75.13±2.16	84.00±2.06	81.49±2.72	71.84±1.87	81.39±1.25	84.27±2.26	83.74±1.14	85.62±1.72	**85.88±2.07**

The observed performance variations across datasets can be attributed to the inherent scene characteristics of each dataset. For instance, the WHU-Hi-LongKou dataset, which features agricultural scenes with regular textural patterns and low spectral variability, is predominantly influenced by spatial-frequency features in classification outcomes. In contrast, the Houston 2013 dataset, representing a complex urban environment, requires the simultaneous modeling of both spectral discrepancies (e.g., variations in material reflectance) and spatial configurations, thereby highlighting the advantage of the HyperMamba's dual-branch collaborative modeling mechanism. Overall, our WD-SSMamba network demonstrates superior accuracy, stability, and robustness across different datasets and scenarios, reinforcing its advanced nature and broad applicability in HSIC tasks. Figs. 12–15 illustrate the classification maps of the comparative models, showing our model's highest accuracies across four benchmark datasets. We believe that our WD-SSMamba model is significant within the context of specific architecture and module design, which not only focuses on classification accuracy but also particularly emphasizes lightweight design.

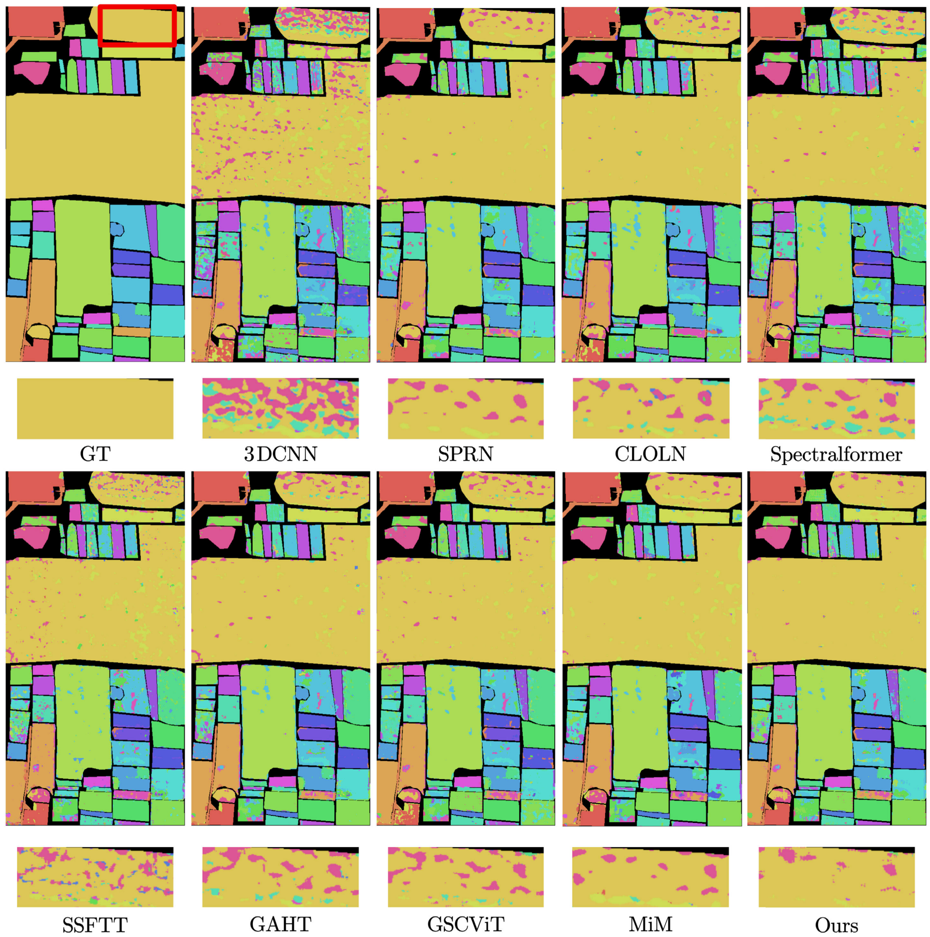


Fig. 14. Classification maps obtained by different models for the WHU-Hi-HongHu dataset. These maps correspond to a single experiment in Table X.

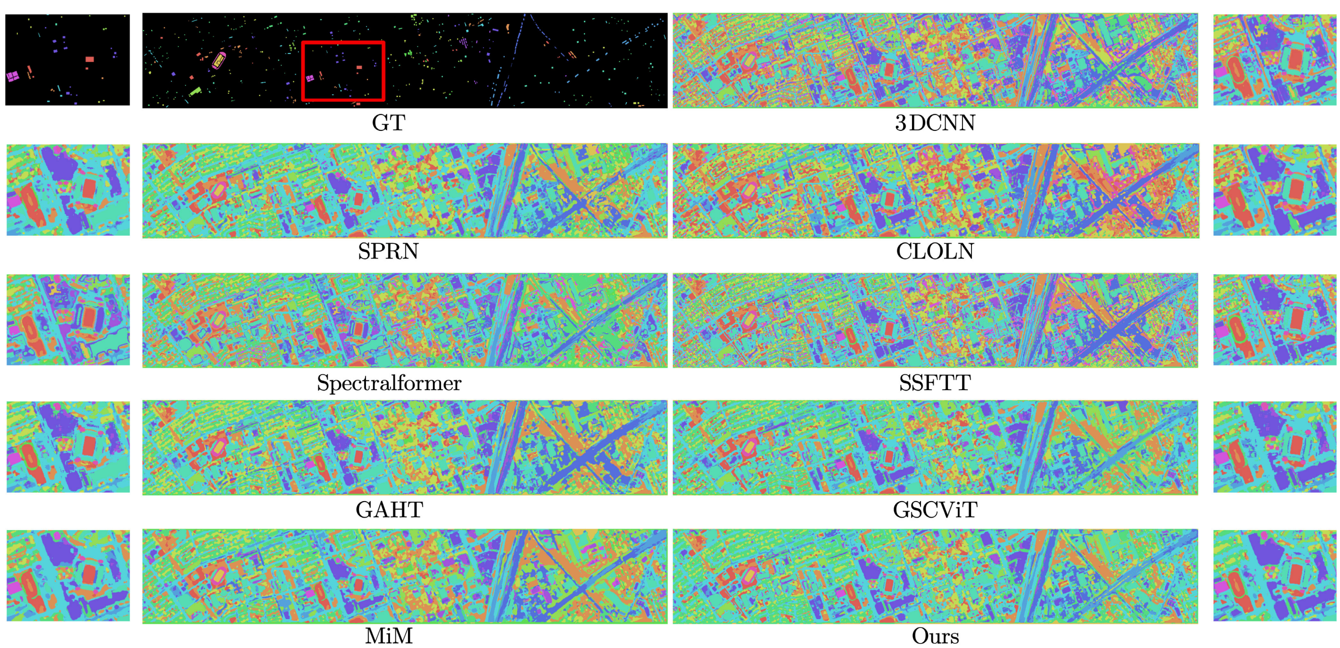


Fig. 15. Classification maps obtained by different models for the Houston 2013 dataset. These maps correspond to a single experiment in Table XI.

G. Analysis on Computational Complexity

In the computational complexity analysis, we evaluated the efficiency of our model using the number of parameters (Params) and FLOPs. As shown in Table XII, our model significantly reduces both Params and FLOPs while maintaining superior classification accuracy compared with mainstream models. For instance, in the WHU-Hi-HongHu dataset, our model uses only 45.50k parameters and 3.02M

TABLE XII
PERFORMANCE COMPARISON OF NINE MODELS ON FOUR HSI DATASETS. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD, AND THE SECOND-BEST RESULTS ARE UNDERLINED

	Model	Params/K	FLOPs/M	FPS	Train Time/s
	3DCNN	101.18	23.26	**1226.69**	**20.1**
	SPRN	178.78	11.57	252.55	97.9
	CLOLN	**3.52**	**0.23**	678.56	36.4
	SF	167.27	14.73	219.83	112.5
PU	SSFTT	484.41	8.44	486.09	50.8
	GAHT	927.11	59.31	225.92	109.5
	GSC-ViT	130.31	7.51	206.87	119.5
	MiM	122.32	132.76	114.23	216.3
	Ours	33.97	2.33	291.29	84.9
	3DCNN	201.98	60.90	**1205.13**	**29.5**
	SPRN	184.88	11.96	288.79	123.1
	CLOLN	**6.19**	**0.40**	575.19	61.8
	SF	543.52	55.80	195.53	181.8
WHULK	SSFTT	1253.95	22.24	558.17	63.7
	GAHT	1514.12	96.84	214.77	165.5
	GSC-ViT	173.06	10.25	211.45	168.0
	MiM	196.46	141.59	118.12	300.9
	Ours	44.65	3.02	278.98	127.4
	3DCNN	449.52	61.15	**1197.86**	**32.5**
	SPRN	186.56	11.96	261.39	149.0
	CLOLN	**6.40**	**0.40**	657.25	59.2
	SF	544.36	55.83	190.00	204.9
WHUHH	SSFTT	1254.79	22.24	636.49	61.2
	GAHT	1514.96	96.84	228.16	170.7
	GSC-ViT	173.91	10.25	211.87	183.7
	MiM	197.31	141.59	117.40	331.6
	Ours	45.50	3.02	276.24	140.9
	3DCNN	190.23	32.43	**1283.28**	**25.8**
	SPRN	181.18	11.67	237.95	139.1
	CLOLN	**4.27**	**0.27**	642.57	51.5
	SF	229.04	22.83	214.48	154.4
Houston	SSFTT	673.73	11.83	617.59	53.6
	GAHT	798.47	51.05	222.65	148.6
	GSC-ViT	141.19	8.19	205.29	161.2
	MiM	135.74	134.51	115.36	287.0
	Ours	36.98	2.50	270.59	122.3

FLOPs, which is much lower than more complex models like SSFTT (1254.79k parameters and 22.24M FLOPs) and GAHT (1514.96k parameters and 96.84M FLOPs). Although our model's Params and FLOPs are higher than those of the CLOLN model, this increase in computational resources is justified by the significant improvement in accuracy. Specifically, while CLOLN achieves extreme lightweight computational complexity, its classification accuracy is lower than our WD-SSMamba model. Moreover, our model achieves a higher FPS compared with most models. Although certain models exhibit higher FPS values, our model consistently demonstrates significantly superior accuracy, ensuring that no tradeoff is made between speed and precision. Overall, our model achieves an excellent balance between computational resources and classification accuracy, demonstrating substantial potential for real-world HSIC tasks.

V. CONCLUSION AND FUTURE RESEARCH

Our proposed WD-SSMamba network demonstrates significant advantages in key metrics based on experimental results. Across four HSI datasets, our model achieves the highest OA, showcasing exceptional classification ability and strong generalization, even with limited samples. Ablation studies have confirmed the crucial roles of the spatial Mamba and spectral Mamba branches in modeling long-range dependencies. In addition, the joint of spectral and spatial frequency-domain information in the FE block improves classification accuracy. Through experiments, we identify the optimal configuration balancing classification accuracy and computational efficiency. Overall, the proposed model exhibits outstanding performance in HSIC tasks, holding broad potential for practical applications. While the WD-SSMamba demonstrates superior performance in terms of classification accuracy and computational efficiency, we acknowledge that there remain limitations in few-shot learning scenarios. In addition, there is untapped potential for further optimization of frequency-domain FE. Future research directions could explore the integration of cross-domain learning paradigms with advanced few-shot classification techniques, thereby enhancing the model's adaptability and robustness across diverse application scenarios.

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